NAVEEN KUMAR REDDY SOMIREDDY

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Research CV · PhD applications

PROFILE

Early-career machine learning researcher focused on federated learning, continual learning, and machine unlearning. Translates research ideas into working algorithms, builds experimental frameworks and baselines, designs controlled experiments, and automates large-scale evaluations, across both the implementation and the theoretical development of the methods. Co-author of FedQCL (CoLLAs 2026, oral presentation, top 8% of accepted papers) and DUET-CLU (under review, AISTATS 2027). Seeking a PhD to develop reliable learning systems that adapt, retain, and forget, through algorithmic innovation, rigorous empirical analysis, and principled optimization.

EDUCATION

Indian Institute of Engineering Science and Technology (IIEST), Shibpur **2021–2025**

B.Tech. in Information Technology | CGPA: 8.68/10

Relevant coursework: Machine Learning, Artificial Intelligence, Data Structures and Algorithms, Linear Algebra, Operating Systems, DBMS, Computer Networks, Computer Architecture

PUBLICATIONS

[C1] Federated Continual Learning as a Distributed Drift-Plus-Penalty Control Problem **CoLLAs 2026 (Oral)**

Nazreen Shah, **Naveen Kumar Reddy Somireddy**, Zubair Shaban, Ranjitha Prasad, Bharath B. N. Accepted; oral presentation (top 8% of accepted papers). Also accepted at the NeurIPS 2026 WiML Workshop. arXiv:2608.21539

PREPRINTS & MANUSCRIPTS

Unified Continual Learning and Unlearning with Adaptive Constraints (DUET-CLU) **Under review · AISTATS 2027**

Nazreen Shah, Prashant Khanduri, **Naveen Kumar Reddy Somireddy**, Bharath B. N., Ranjitha Prasad.

FedResilience: Concurrent Handling of Client Dropout and Slow Learners in Federated Learning **Manuscript**

Naveen Kumar Reddy Somireddy (first author), Sachin Dudda Nagaraju, Annappa B. Proximity-based weight transfer and adaptive batch scheduling.

RESEARCH EXPERIENCE

Research Assistant, Federated Continual Learning & Unlearning **Dec 2024–Mar 2026**

Indraprastha Institute of Information Technology Delhi (IIIT Delhi) | Advisor: Prof. Ranjitha Prasad

FedQCL: Federated Continual Learning via Drift-Plus-Penalty **Co-author · CoLLAs 2026 (Oral)**

- Studied catastrophic forgetting and client drift by formulating federated continual learning as a stochastic control problem. FedQCL uses client-local virtual queues to track accumulated forgetting and a tunable parameter to balance stability and plasticity, without gradient projection or extra communication.
- With ResNet-18 and five non-IID clients, reported 54.5% vs. 51.0% for DER++-FL on Split-CIFAR-100, 85.7% vs. 83.9% on Split-CIFAR-10, and 33.4% vs. 31.0% on Split-TinyImageNet (Task-IL). Class-IL on Split-CIFAR-100: 24.92% vs. 24.32% (near-IID) and 14.66% vs. 14.02% (non-IID) against DER++.

My contributions

- Implemented the FedQCL method and its full experimental codebase; implemented or integrated all baselines (FedAvg, FedProto, EWC-FL, LwF-FL, ER-FL, DER-FL, DER++-FL, CCVR, FOT, BI); contributed to the theoretical development of the framework.
- Designed experiments and ablations, tuned hyperparameters, and ran the evaluation suite across data heterogeneity, buffer size, client count, and task count in Task-IL and Class-IL settings; produced the results reported in the paper. Code: github.com/Naveensomireddy4/FedQCL_Task · github.com/Naveensomireddy4/FedQCL_CIL

DUET-CLU: Unified Continual Learning and Unlearning **Co-author · Jul–Oct 2026 · AISTATS 2027 (under review)**

- Formulates retention and non-resurgence as task-level constraints in a primal-dual framework with adaptive slack variables and dual multipliers; includes a dual-sensitivity characterization and a slack-dual equilibrium under convex per-request assumptions.

My contributions

- Implemented the DUET-CLU framework and its experimental codebase; implemented or integrated the comparison baselines (Retrain, EWC, LwF, ER, DER++, LSF, CLPU, CLMUL, UnCLe, PALL); contributed to the theoretical formulation and analysis.
- Designed experiments and ablations, tuned hyperparameters, and evaluated the effects of slack variables and dual multipliers on continual learning and unlearning in Task-IL and Class-IL settings.

Continual Learning for Federated Speech Models (Whisper-small)

- Built a federated training pipeline for Whisper-small and integrated A-GEM as the continual-learning method; reported 96% accuracy (IID), nearly matching centralized training on all the data, and 80% (non-IID).

Research Intern, Federated Learning **May 2024–Nov 2024**

National Institute of Technology Karnataka (NITK), Surathkal | Advisor: Prof. Annappa B.

FedResilience: Client Dropout and Slow-Learner Resilience **First author**

- Proposed a FedAvg-based framework combining proximity-based weight transfer from the most similar active client (cosine similarity) for dropout handling with adaptive dual-batch scheduling for slow learners.
- On BloodMNIST with ResNet-18 under non-IID data, reached 82.1% accuracy in 8 rounds, versus 80.95% for FedAvg in 15 rounds.

INDUSTRY EXPERIENCE

Data Analyst & Data Engineer (MLOps), Tredence, Bengaluru **Feb 2026–Oct 2026**

- Migrated five data pipelines (20 tables; ~3 billion records each) from Hive to GCP, converted SQL jobs to Spark (Scala), automated Dataproc runs, and moved reporting to Power BI.

SELECTED PROJECT

Real-Time Scene Graph Generation **Jan 2024–Mar 2024**

YOLOv5, PAN-FPN, GloVe, PyTorch | Advisor: Dr. Arindam Biswas

- Built a real-time scene-graph model for object detection and relation classification using PAN-FPN feature fusion, spatial encodings, and ROI Align; 14.5 mR@100 at 30 FPS on the Panoptic Scene Graph dataset.

TECHNICAL SKILLS

Research: Federated learning, continual learning, machine unlearning, constrained optimization, primal-dual methods, experiment design and automation

Programming & ML: Python, C++, SQL, PyTorch, TensorFlow, Hugging Face

Data & infrastructure: Hive, Spark, Scala, GCP, Power BI, MLOps, Docker, Git, Linux

Models & benchmarks: ResNet-18, Whisper; Split-CIFAR-10/100, Split-TinyImageNet, BloodMNIST

Baseline library: FedAvg, FedProto, EWC, LwF, ER, DER/DER++, A-GEM, CCVR, FOT, BI, LSF, CLPU, CLMUL, UnCLe, PALL (github.com/Naveensomireddy4/Baselines)

HONORS

- **CoLLAs 2026:** Oral presentation; top 8% of accepted papers.
- **LeetCode:** Knight (top 3%); 1,100+ problems solved; Rank 97 in Biweekly Contest 133.

REFERENCES

Prof. Ranjitha Prasad (IIIT Delhi) · Prof. Annappa B. (NITK Surathkal). Contact details available on request.